ACROCARE – CENTRE FOR HAPPINESS & WELL-BEING

CMHSI SOFTWARE DEVELOPMENT SPECIFICATION

Questionnaire, Scoring Engine, Critical-Safety Workflow & Automated Reporting

1. System Overview

The system should support the following workflow:

1. Collect student information and 54 questionnaire responses.
2. Store the original text response and its corresponding numeric score.
3. Calculate the nine domain scores automatically.
4. Calculate the combined descriptive score.
5. Assign sample-referenced relative bands where applicable.
6. Run a separate critical-item safety check.
7. Generate an internal counsellor/admin dashboard.
8. Generate a standardized individual student report/PDF.
9. Maintain follow-up, safety-review, referral, and professional-note records.

2. Core Questionnaire Configuration

Configuration	Required specification
Instrument	CMHSI – AcroCare psychometric screening instrument
Total items	54
Domains	9
Items per domain	6
Response scale	Never, Rarely, Sometimes, Often, Almost Always
Numeric range	0–4 per item
Reverse scoring	None
Domain score range	0–24
Combined score range	0–216
Primary interpretation	Domain-wise profile
Combined score	Descriptive only; not the primary interpretation

3. Response-to-Score Conversion

Response option	Numeric score
Never	0
Rarely	1
Sometimes	2
Often	3
Almost Always	4

Implementation rule: Store both the original response text and the converted numeric value. No item is reverse scored.

4. Domain Mapping / Scoring Algorithm

The following mapping must be hard-coded or stored as a controlled master configuration:

Domain	Code	Items/Questions	Maximum
Anxiety	AN	1, 10, 19, 28, 37, 46	24
Depression	DP	2, 11, 20, 29, 38, 47	24
Suicidal Ideation	SI	3, 12, 21, 30, 39, 48	24
Substance Abuse	SA	4, 13, 22, 31, 40, 49	24
Self-Esteem Problems	SE	5, 14, 23, 32, 41, 50	24
Interpersonal Problems	IP	6, 15, 24, 33, 42, 51	24
Family Problems	FP	7, 16, 25, 34, 43, 52	24
Academic Stress	AS	8, 17, 26, 35, 44, 53	24
Career Concerns	CC	9, 18, 27, 36, 45, 54	24

5. Score Calculation Rules

- Convert each response using the response-to-score table.
- For each domain, sum the six item scores assigned to that domain.
- Domain Score = sum of six domain item scores.
- Each domain has a maximum score of 24.
- Combined Score = sum of all 54 numeric item scores.
- Combined score has a possible range of 0–216.
- The domain profile must remain the primary basis for interpretation.
- The combined score must be labelled descriptive only.

Example: If Anxiety responses are Rarely, Never, Rarely, Rarely, Sometimes, Sometimes, the numeric conversion is 1 + 0 + 1 + 1 + 2 + 2 = 7, therefore Anxiety = 7/24.

6. Relative-Band Algorithm

These bands are sample-referenced relative bands based on N = 603. They are not clinical severity thresholds, diagnostic cut-offs, or population norms.

Domain	Score range	Band
Anxiety	0–5	Lower Relative
Anxiety	6–9	Mild Relative
Anxiety	10–14	Moderate Relative
Anxiety	15–24	Higher Relative
Depression	0–4	Lower Relative
Depression	5–8	Mild Relative
Depression	9–14	Moderate Relative
Depression	15–24	Higher Relative
Suicidal Ideation	0	Lower Relative
Suicidal Ideation	1–2	Mild Relative
Suicidal Ideation	3–5	Moderate Relative
Suicidal Ideation	6–24	Higher Relative
Self-Esteem Problems	0–2	Lower Relative
Self-Esteem Problems	3–6	Mild Relative
Self-Esteem Problems	7–11	Moderate Relative
Self-Esteem Problems	12–24	Higher Relative
Interpersonal Problems	0–4	Lower Relative
Interpersonal Problems	5–8	Mild Relative
Interpersonal Problems	9–13	Moderate Relative
Interpersonal Problems	14–24	Higher Relative
Substance Abuse	0–4	Lower Relative
Substance Abuse	5–9	Mild Relative
Substance Abuse	10–15	Moderate Relative
Substance Abuse	16–24	Higher Relative
Family Problems	0–2	Lower Relative
Family Problems	3–6	Mild Relative
Family Problems	7–10	Moderate Relative
Family Problems	11–24	Higher Relative
Academic Stress	0–4	Lower Relative
Academic Stress	5–8	Mild Relative
Academic Stress	9–13	Moderate Relative
Academic Stress	14–24	Higher Relative
Career Concerns	0–4	Lower Relative
Career Concerns	5–9	Mild Relative
Career Concerns	10–15	Moderate Relative
Career Concerns	16–24	Higher Relative

7. Developer Acceptance Tests

Before launch, test the system using at least 5–10 dummy students and compare automated output with manual calculations.

Test	Input	Expected result
Test 1: All Never	All 54 items = 0	Every domain = 0; Combined = 0; No critical flag
Test 2: All Almost Always	All 54 items = 4	Every domain = 24; Combined = 216; Items 21, 30, 39 flagged
Test 3: Dummy test case	Use domain values below	AN 7; DP 7; SI 5; SA 4; SE 4; IP 4; FP 11; AS 7; CC 17; Combined 66; Item 39 flag
Test 4: Low total + critical item	All items = 0 except Item 39 = 1	Combined = 1/216; SI = 1/24; Critical flag = YES

8. Dummy Test Case – Detailed Values

Domain	Six numeric responses	Calculation	Score
Anxiety	1, 0, 1, 1, 2, 2	1+0+1+1+2+2	7/24
Depression	1, 0, 1, 1, 2, 2	1+0+1+1+2+2	7/24
Suicidal Ideation	1, 0, 0, 0, 2, 2	1+0+0+0+2+2	5/24
Substance Abuse	0, 0, 0, 0, 2, 2	0+0+0+0+2+2	4/24
Self-Esteem	0, 0, 0, 0, 2, 2	0+0+0+0+2+2	4/24
Interpersonal	0, 0, 0, 0, 2, 2	0+0+0+0+2+2	4/24
Family	2, 0, 2, 2, 2, 3	2+0+2+2+2+3	11/24
Academic	1, 0, 1, 0, 2, 3	1+0+1+0+2+3	7/24
Career	3, 2, 4, 3, 2, 3	3+2+4+3+2+3	17/24
Combined	—	Sum of all 54 scores	66/216

Critical-item check for this test case: Item 21 = 0; Item 30 = 0; Item 39 = 2. Expected result: Critical safety flag = YES and Safety Review Required.

9. Recommended Software Architecture

Keep the software in three separate layers:

Layer	Contents
1. RAW DATA	54 original responses; student information; administration metadata
2. SCORING ENGINE	Text-to-numeric conversion; item-to-domain mapping; domain scores; bands; critical-item logic
3. REPORT ENGINE	Scores; interpretations; priority areas; charts; follow-up section; PDF generation

This separation will allow AcroCare to change report design later without accidentally changing the scoring algorithm.

10. Domain-wise recommendations based on Lower, Mild, Moderate and Higher Relative bands

1. Anxiety (AN)

Band / situation	Automated recommendation
Lower Relative	Continue healthy routines, adequate sleep, relaxation practices, and supportive social connections. Encourage awareness of situations that contribute to worry.
Mild Relative	Encourage slow breathing, grounding exercises, structured routines, and identification of worry triggers. Psychoeducation and optional counselling may be offered if concerns persist.
Moderate Relative	Recommend a counsellor discussion to explore the nature, frequency, and impact of anxiety. Encourage coping strategies, relaxation practice, sleep regulation, and monitoring of symptoms.
Higher Relative	Recommend timely professional review to understand the intensity and functional impact of anxiety. Develop an individualized support plan addressing coping skills, sleep, daily functioning, and stressors.

2. Depression (DP)

Band / situation	Automated recommendation
Lower Relative	Encourage meaningful activities, healthy routines, supportive relationships, and regular self-care.
Mild Relative	Encourage daily routines, appropriate physical activity, positive social engagement, enjoyable activities, and awareness of mood changes. Offer psychoeducation and optional counselling.
Moderate Relative	Recommend a counsellor discussion to explore mood, energy, concentration, interest in activities, and impact on daily functioning. Encourage structured activity planning and social support.
Higher Relative	Recommend timely professional assessment and individualized support to explore persistent mood-related concerns and functional impact. Offer counselling or referral based on professional review.

4. Suicidal Ideation (SI)

Band / situation	Automated recommendation
Lower Relative	Encourage emotional support, healthy coping strategies, and awareness of available campus counselling resources. Maintain well-being check-ins where appropriate.
Mild Relative	Recommend a supportive counsellor conversation to understand the context of reported thoughts, emotional distress, and available support resources. Encourage help-seeking and monitoring.
Moderate Relative	Recommend prompt professional review to explore the nature, frequency, context, and impact of suicidal thoughts. Assess safety and support needs directly.
Higher Relative	Recommend urgent professional safety assessment and immediate follow-up. The appropriate response should be determined through direct assessment and institutional protocol.

5. Substance Abuse (SA)

Band / situation	Automated recommendation
Any reported concern	Encourage a confidential, non-judgmental counsellor discussion to understand the nature, frequency, context, and functional impact of substance use. Provide psychoeducation on healthier coping strategies.
Use affects functioning	Recommend professional review and individualized support to explore patterns, triggers, consequences, and available referral options.
Student wants to reduce/stop	Offer supportive guidance, motivational conversations, and referral to an appropriate qualified professional or de-addiction service where indicated.

6. Self-Esteem Problems (SE)

Band / situation	Automated recommendation
Lower Relative	Encourage self-awareness, recognition of strengths, healthy self-talk, and meaningful activities.
Mild Relative	Encourage identifying strengths, challenging negative self-judgments, setting realistic goals, and accepting constructive feedback.
Moderate Relative	Recommend counselling support to explore negative self-perceptions, self-comparison, confidence concerns, and their impact on functioning.
Higher Relative	Recommend individualized professional support to explore persistent self-worth concerns and negative self-beliefs. Encourage gradual strength-building and supportive goal-setting.

7. Interpersonal Problems (IP)

Band / situation	Automated recommendation
Lower Relative	Encourage healthy relationships, open communication, mutual respect, and positive social activities.
Mild Relative	Encourage active listening, assertive communication, boundary-setting, and seeking supportive connections.
Moderate Relative	Recommend a counsellor discussion to explore relationship difficulties, loneliness, trust concerns, communication patterns, and their impact on well-being.
Higher Relative	Recommend individualized counselling support for persistent interpersonal concerns, social isolation, trust difficulties, or relationship-related distress.

8. Family Problems (FP)

Band / situation	Automated recommendation
Lower Relative	Encourage open family communication, mutual understanding, supportive relationships, and healthy personal boundaries.
Mild Relative	Encourage constructive communication, identifying manageable sources of family stress, and seeking support from trusted individuals.
Moderate Relative	Recommend a counsellor discussion to explore family stress, expectations, communication difficulties, and their impact on emotional well-being.
Higher Relative	Recommend individualized professional support to understand persistent family-related distress and identify safe, practical coping and support options. Consider family involvement only when appropriate.

9. Academic Stress (AS)

Band / situation	Automated recommendation
Lower Relative	Encourage effective study routines, realistic academic goals, balanced schedules, and healthy breaks.
Mild Relative	Encourage time management, prioritization, planned study sessions, short breaks, and academic clarification or support when required.

Moderate Relative	Recommend a counsellor discussion to explore workload, expectations, performance concerns, and time-management difficulties. Encourage a structured academic action plan.
Higher Relative	Recommend timely professional and academic support to understand the impact of academic stress. Develop an individualized plan addressing workload, prioritization, expectations, and institutional resources.

10. Career Concerns (CC)

Band / situation	Automated recommendation
Lower Relative	Encourage career exploration, awareness of interests and strengths, and gradual planning for future goals.
Mild Relative	Encourage exploring interests, gathering reliable information, identifying strengths, and discussing options with a career counsellor, mentor, or faculty member.
Moderate Relative	Recommend structured career guidance to explore uncertainty, decision-making difficulties, interests, abilities, and realistic options. Break decisions into manageable steps.
Higher Relative	Recommend timely individualized career counselling to explore persistent career-related distress, decision pressure, uncertainty, and impact on functioning. Develop a structured career-planning support plan.

11. Software Implementation Requirement

- Calculate each domain score first.
- Assign the applicable relative band using approved thresholds.
- Retrieve only the recommendation corresponding to the obtained band.
- Display recommendations for all relevant domains, not only the highest-scoring domain.
- Do not assign a relative band to Substance Abuse.
- Run critical-item logic independently from routine recommendations.
- Use non-diagnostic and non-stigmatizing language.
- Allow counsellors to add professional notes or supplementary recommendations after review.

12. Standard Report Introductory Wording

Band	Suggested wording
Lower Relative	This domain does not indicate a relatively elevated screening score in the current reference framework. Continue preventive well-being practices.
Mild Relative	This domain reflects a mildly elevated relative screening score. Practical coping strategies and monitoring may be helpful.
Moderate Relative	This domain reflects a moderately elevated relative screening score. A counsellor discussion is recommended to understand the concern and its impact.
Higher Relative	This domain reflects a higher relative screening score. Timely professional review is recommended for further exploration and individualized support.